

Nebulas

Stars are created deep inside dark clouds of gas and dust known as nebulas. Astronomers measure the brightness of a star using a scale of apparent magnitude. This scale describes how bright a star

is when viewed from Earth. Stars with a magnitude of up to 6 are visible with the naked eye. Those of 7 and above need binoculars or a telescope to be seen. Below are a few of the brightest nebulas.



Name: Carina Nebula
Designation: NGC 3372
Constellation: Carina
Magnitude: 1
Distance: 6,500 light years
Visibility: Naked eye



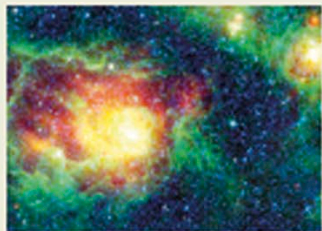
Name: Dumbbell Nebula
Designation: M27
Constellation: Vulpecula
Magnitude: 7.5
Distance: 1,360 light years
Visibility: Binoculars



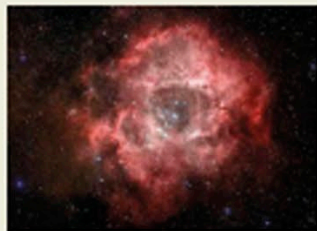
Name: Orion Nebula
Designation: M42
Constellation: Orion
Magnitude: 4
Distance: 1,340 light years
Visibility: Naked eye



Name: Helix Nebula
Designation: NGC 7293
Constellation: Aquarius
Magnitude: 7.6
Distance: 700 light years
Visibility: Binoculars



Name: Lagoon Nebula
Designation: M8
Constellation: Sagittarius
Magnitude: 6
Distance: 4,100 light years
Visibility: Naked eye



Name: Rosette Nebula
Designation: NGC 2237
Constellation: Monoceros
Magnitude: 9
Distance: 5,200 light years
Visibility: Binoculars

Galaxies

Galaxies are huge collections of stars, gas, dust, and dark matter. They started life billions of years ago, slowly forming into distinctive shapes. There are four main types of galaxies—spirals, barred spirals, elliptical, and irregular. Our star, the Sun, lives in a barred-spiral galaxy called the Milky Way. This table lists some of the galaxies that can be seen with binoculars or the naked eye.

KEY



Irregular



Barred spiral



Spiral



Elliptical

Type	Name	Designation	Constellation	Apparent magnitude	Distance	Visibility
	Large Magellanic Cloud	LMC	Dorado/Mensa	0.9	160,000 light years	Naked eye
	Small Magellanic Cloud	SMC	Tucana	2.7	200,000 light years	Naked eye
	Andromeda Galaxy	M32	Andromeda	3.4	2.5 million light years	Naked eye
	Triangulum Galaxy	M33	Triangulum	5.7	2.9 million light years	Binoculars
	Centaurus A	NGC 5128	Centaurus	6.8	13.7 million light years	Binoculars
	Bode's Galaxy	M81	Ursa Major	6.9	11.8 million light years	Binoculars
	Southern Pinwheel	M83	Hydra	7.5	15.2 million light years	Binoculars
	Sculptor Galaxy	NGC 253	Sculptor	8.0	11.4 million light years	Binoculars